

Technical Document #817: Software Engineering - Comprehensive Overview

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Continuous deployment (CD) automatically deploys code changes to production. It enables rapid, reliable releases.

Design patterns provide reusable solutions to common software design problems. Singleton, factory, and observer patterns are widely used.

Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- API design defines how software components communicate.
- Code review examines code changes before merging.
- Domain-driven design (DDD) models software around business domains.